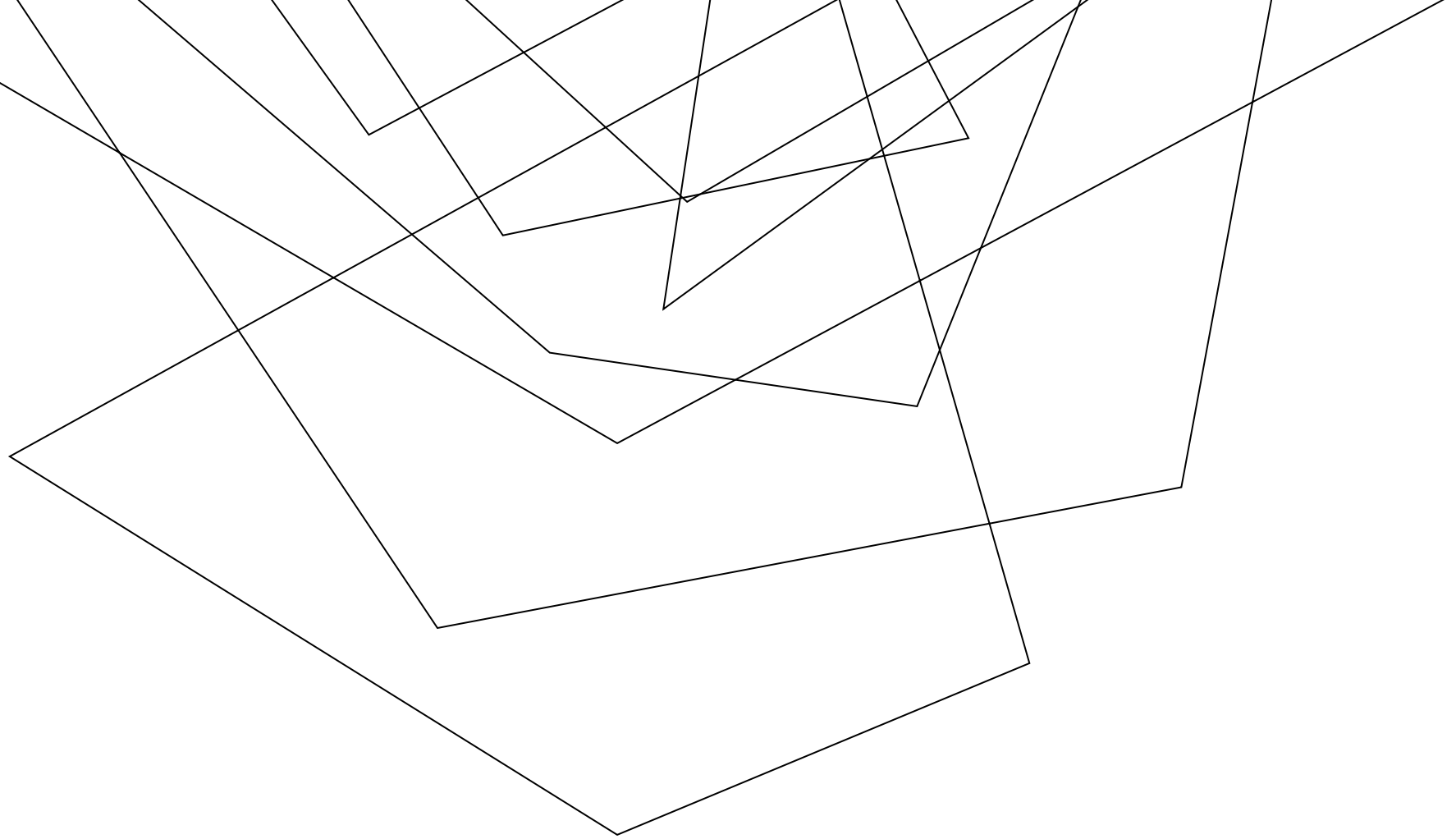
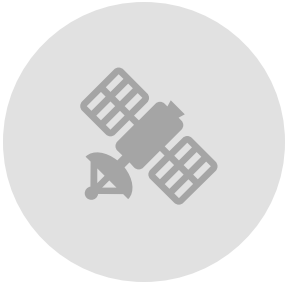


PT7003: CHRONIC PAIN

Irvind Sihota

LEARNING OUTCOMES



Demonstrate an understanding of the transmission system for nociception.



Describe the relationship between pain and tissue damage

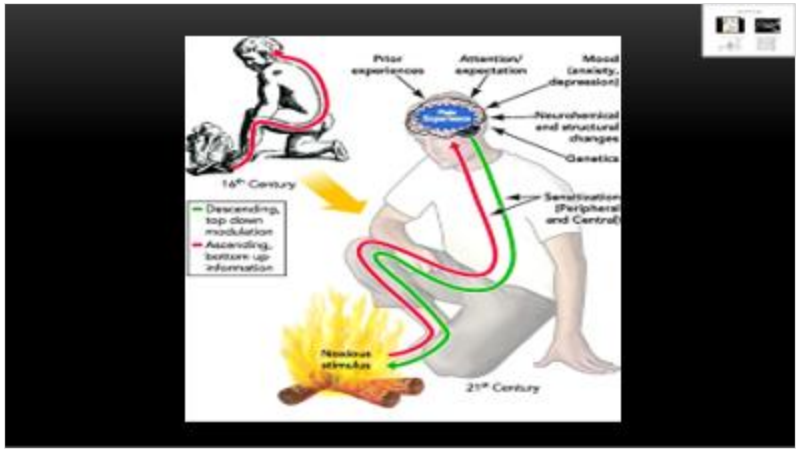


Explain the basic concept of pain as a multisystem body response to danger or damage

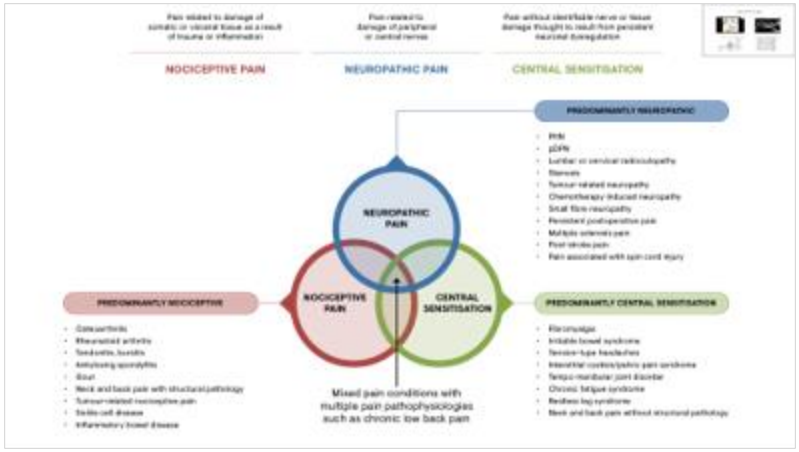


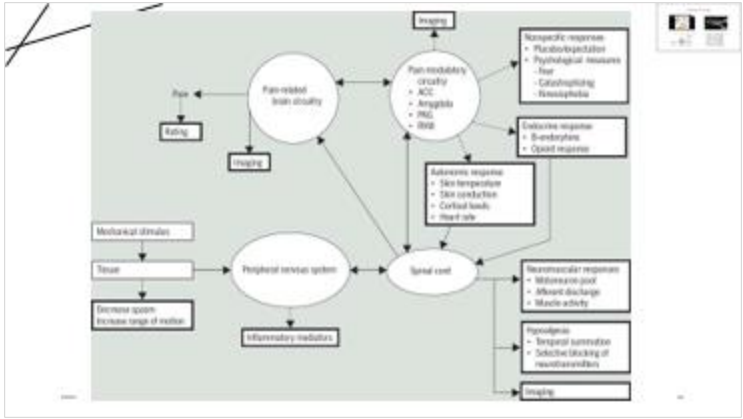
Describe the basic responses to the experience of pain and factors that lead to the development of chronic pain.

IMPORTANT SLIDES









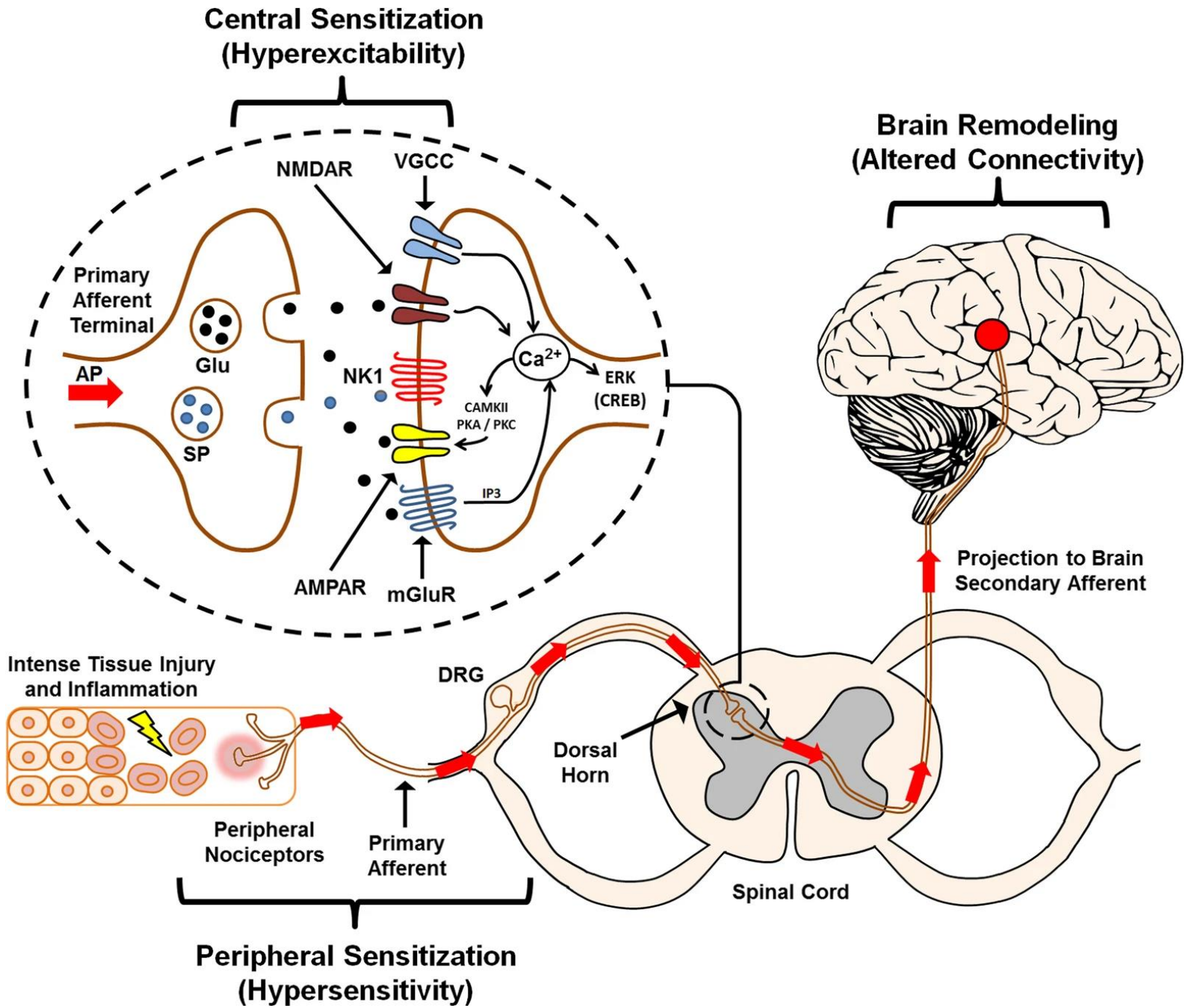
TRANSMISSION SYSTEM FOR NOCICEPTION

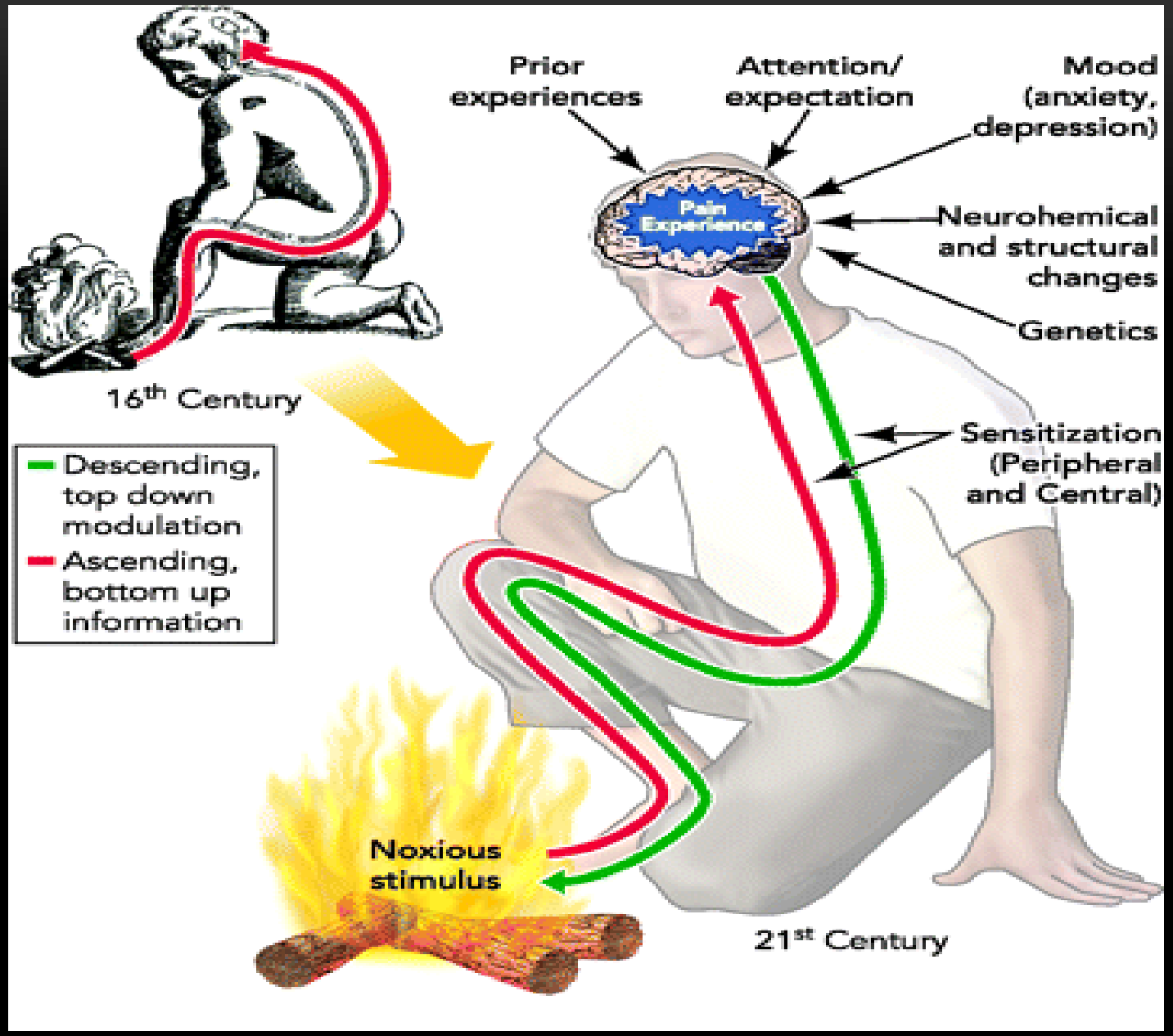
&

RELATIONSHIP BETWEEN PAIN AND TISSUE DAMAGE

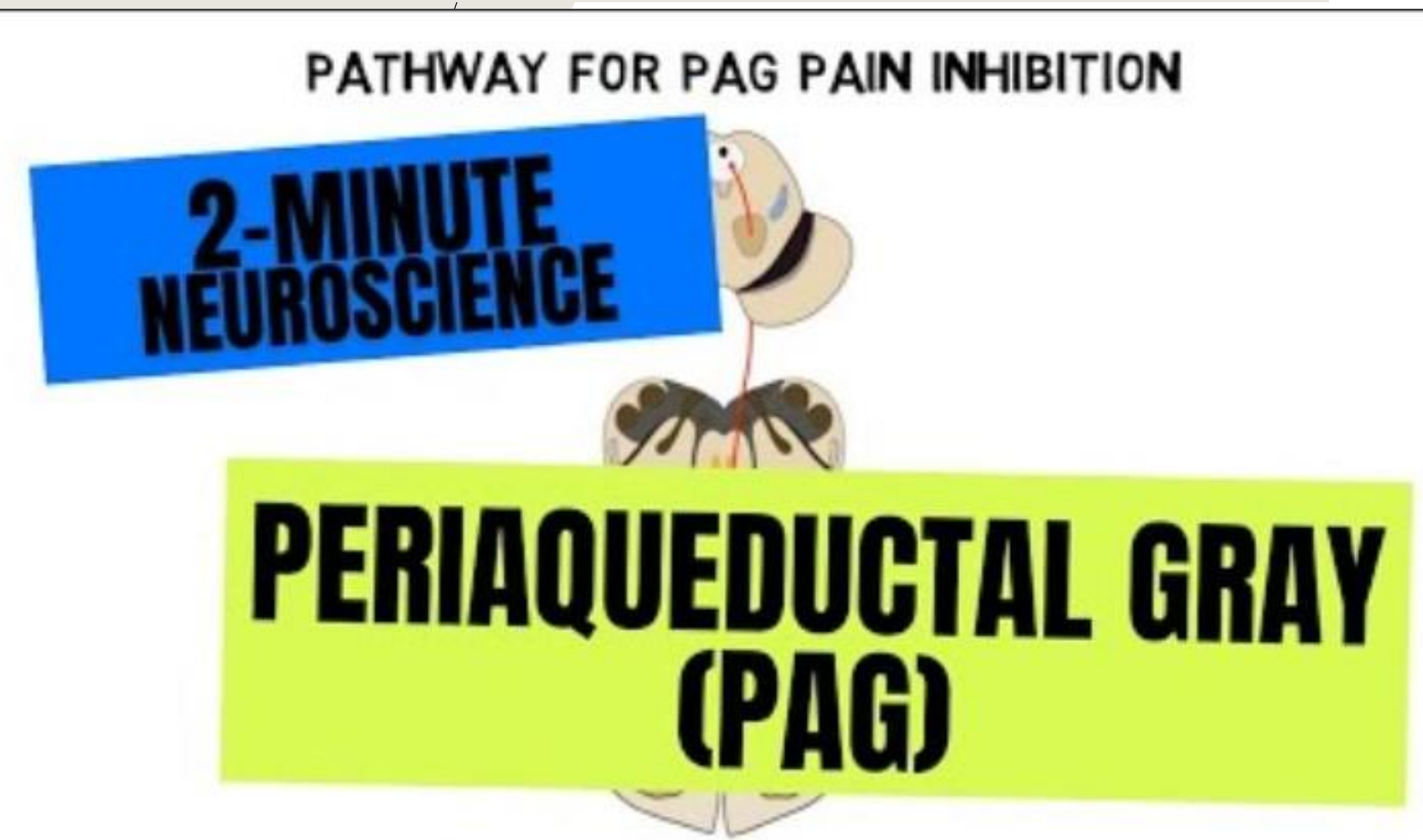
Chronic Pain & Sensitisation



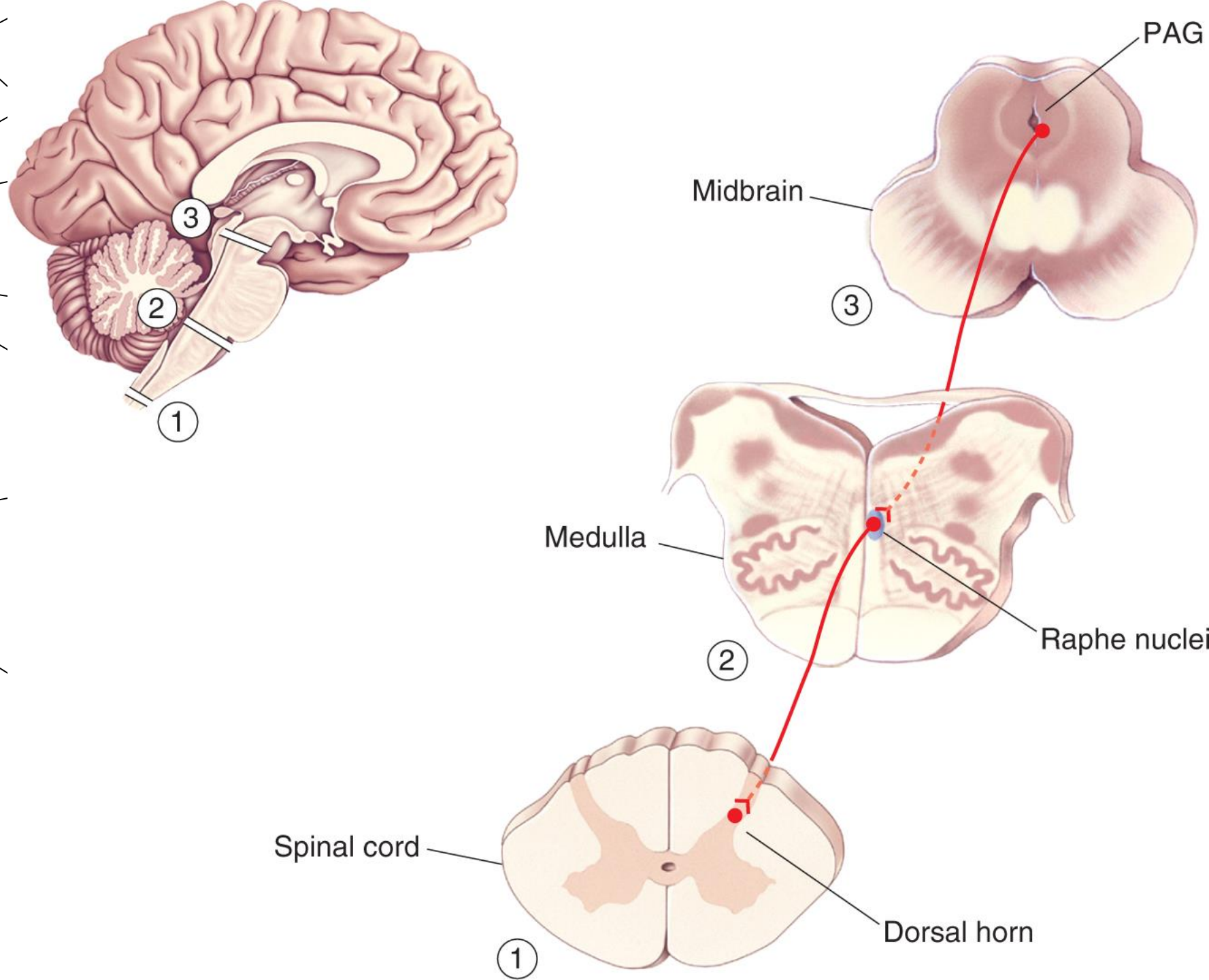

Central Sensitization
(Hyperexcitability)
NMDAR
VGCC
Primary
Afferent
Terminal
AP
Glu
SP
NK1
Ca2+
ERK
(CREB)
CAMKII
PKA / PKC
IP3
AMPAR
mGluR
Brain Remodeling
(Altered Connectivity)
Projection to Brain
Secondary Afferent
Intense Tissue Injury
and Inflammation
DRG
Dorsal
Horn
Peripheral
Nociceptors
Primary
Afferent
Spinal Cord
Peripheral Sensitization
(Hypersensitivity)

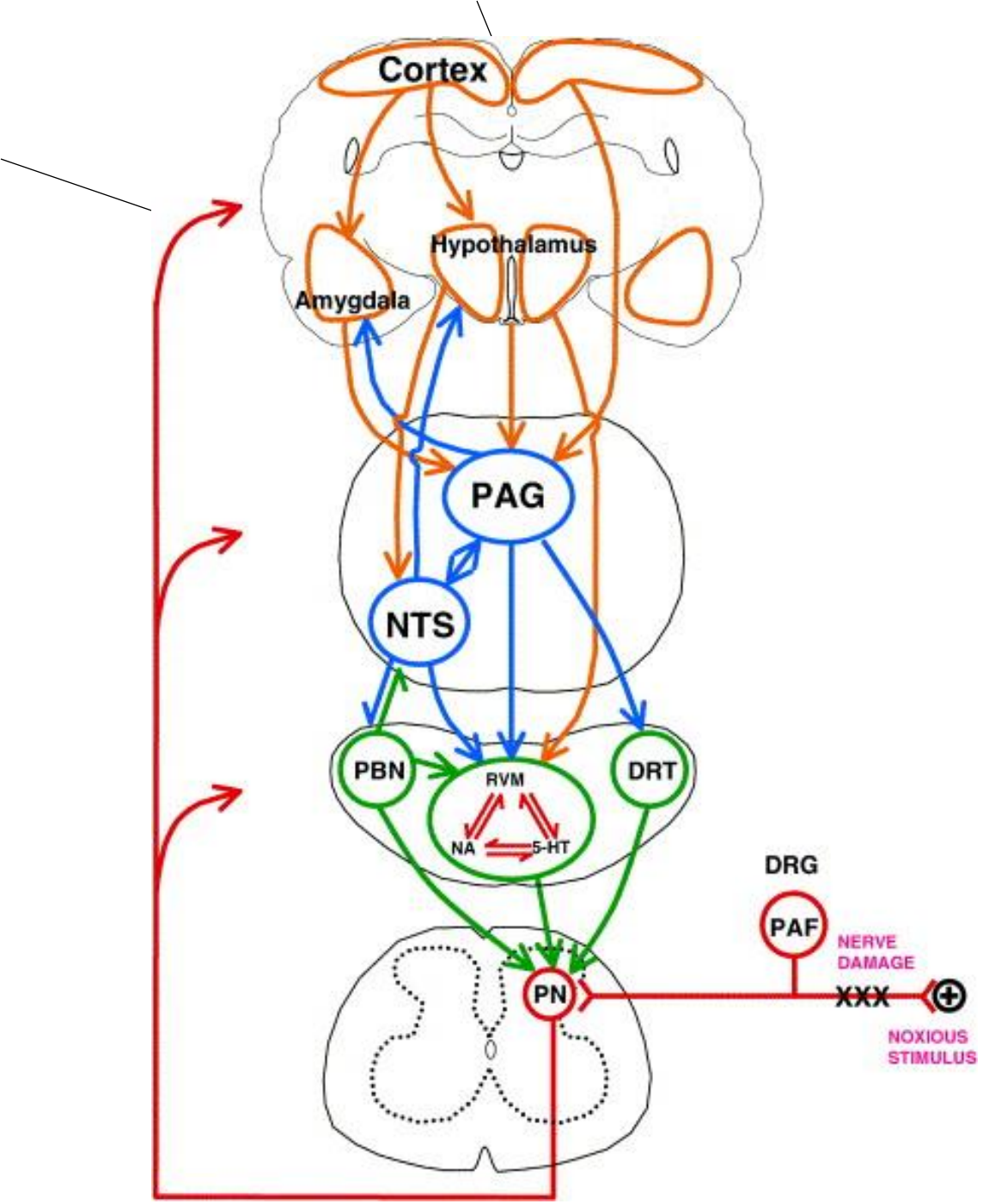

Prior experiences
Attention/ expectation
Mood (anxiety, depression)
Pain Experience
Neurohemical and structural changes
Genetics
Sensitization (Peripheral and Central)
16th Century
Descending, top down modulation
Ascending, bottom up information
Noxious stimulus
21st Century

Periaqueductal Grey



Descending pain-control pathways. A variety of brain structures, many of which are affected by behavioural state, can influence activity within the periaqueductal grey matter (PAG) of the midbrain. The PAG can influence the raphe nuclei of the medulla, which in turn can modulate the flow of nociceptive information through the dorsal horns of the spinal cord.





Descending Control:

Cerebral structures involved in the initiation and modulation of descending controls of nociceptive information. Note the strategic location of the periaqueductal grey (PAG) and the reciprocal nature of many interconnecting pathways.

CX, cortex

NTS, nucleus tractus solitarius;

PBN, parabrachial nucleus;

DRT, dorsoreticular nucleus;

RVM, rostroventral medulla;

NA, noradrenaline

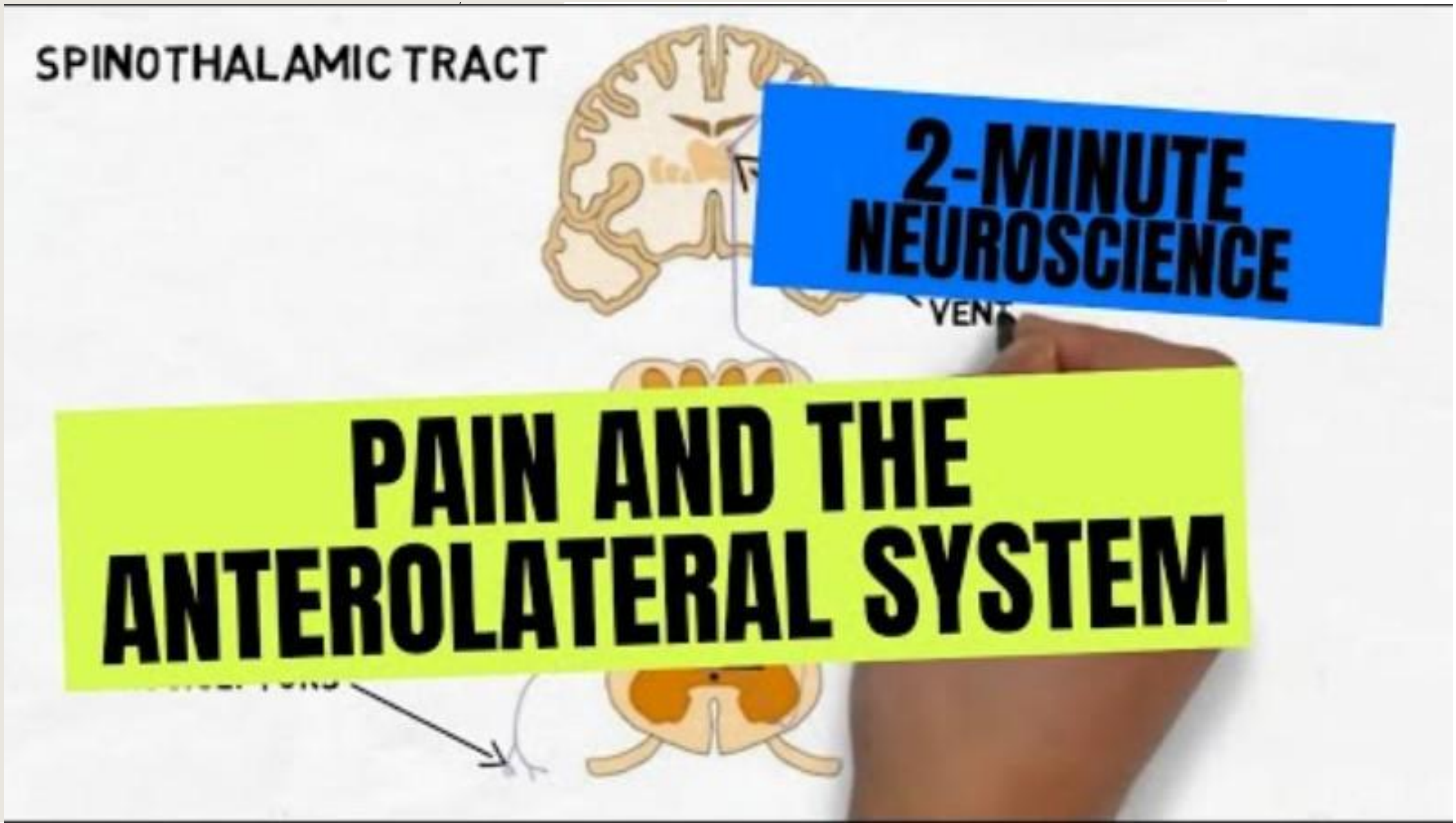
5-HT, serotonergic

PAF, primary afferent fibre and

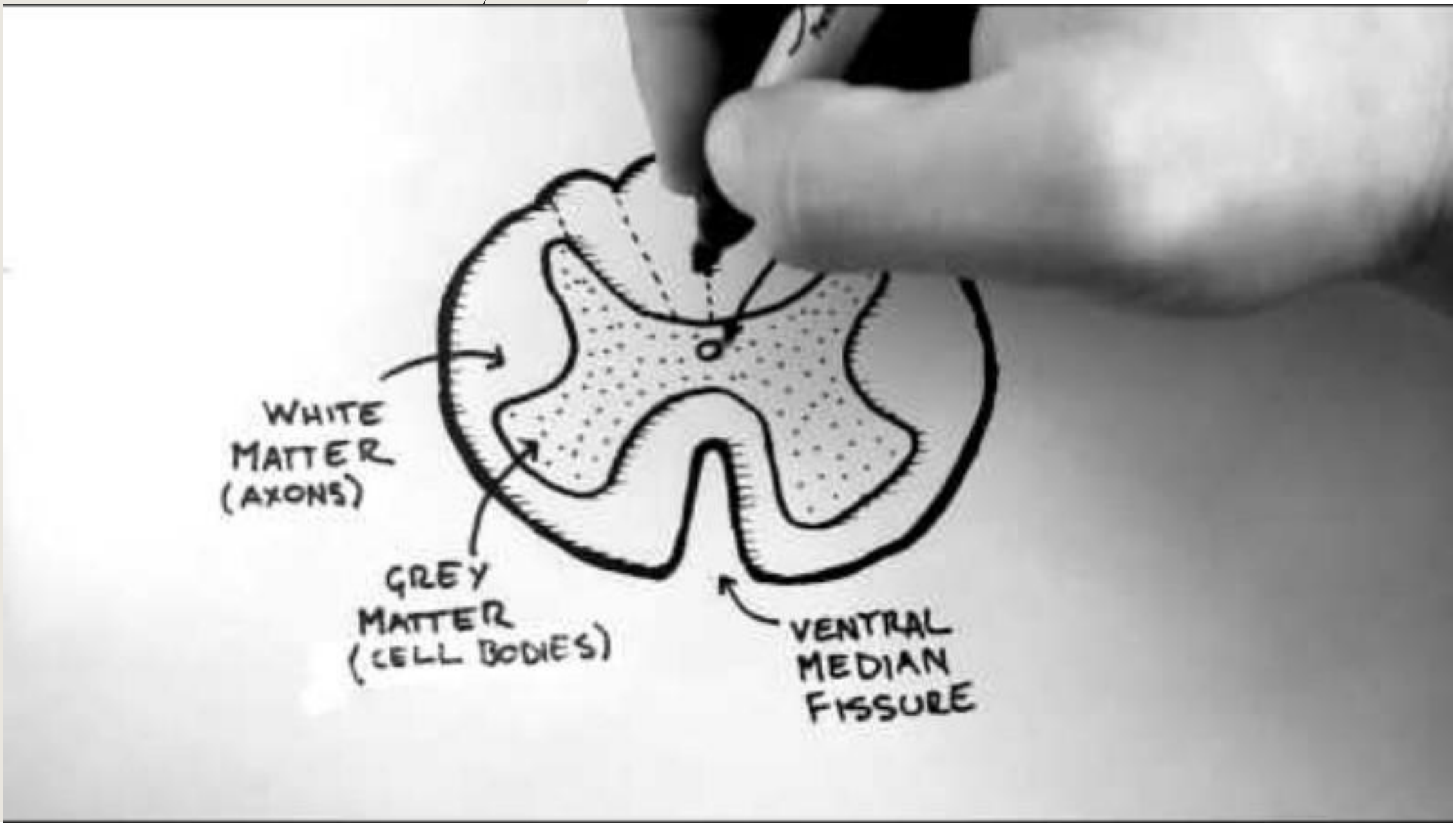
DRG, dorsal root ganglion.

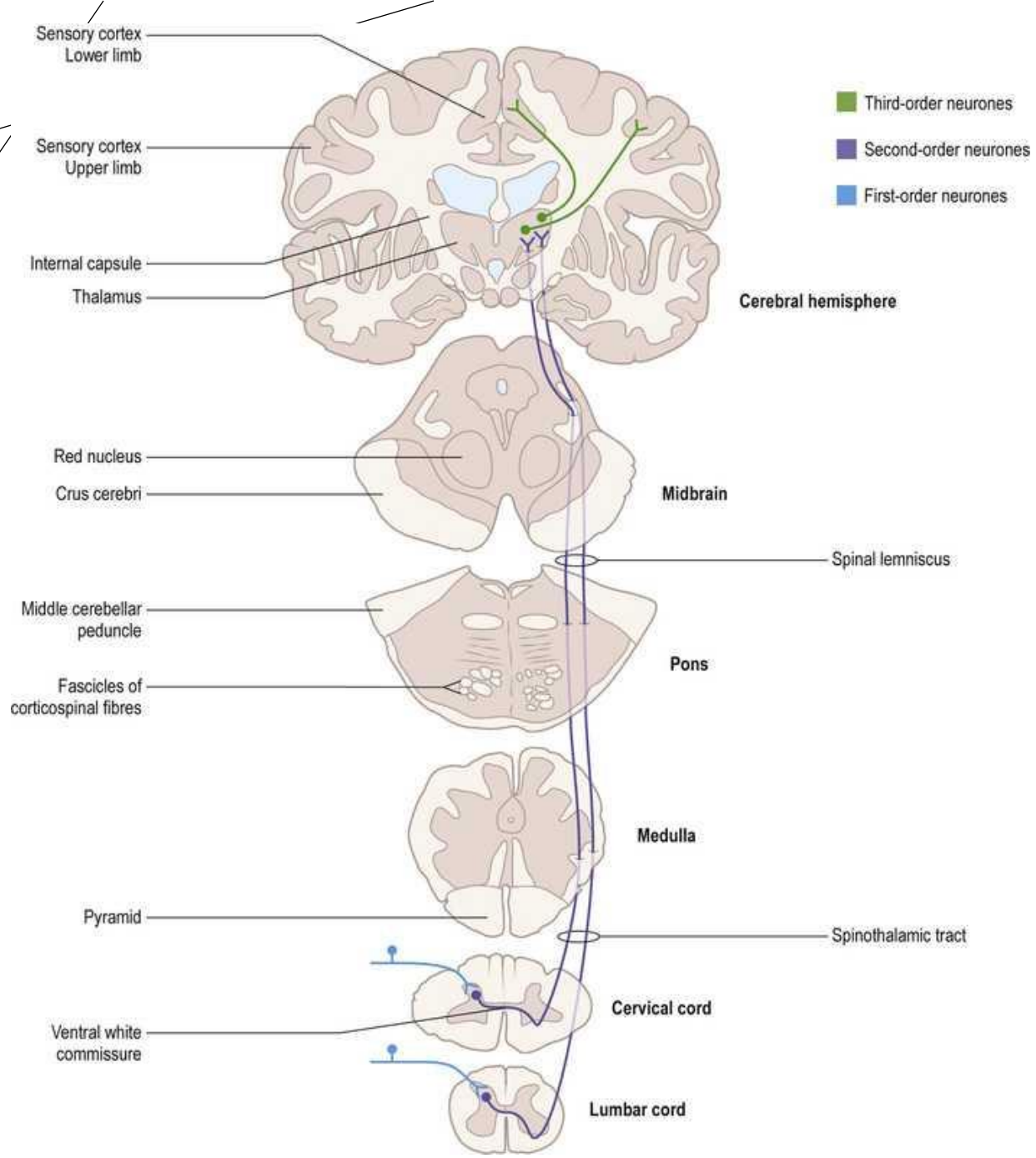
PN - projection neurones (PNs)

Pain – Nociceptors – anterolateral system

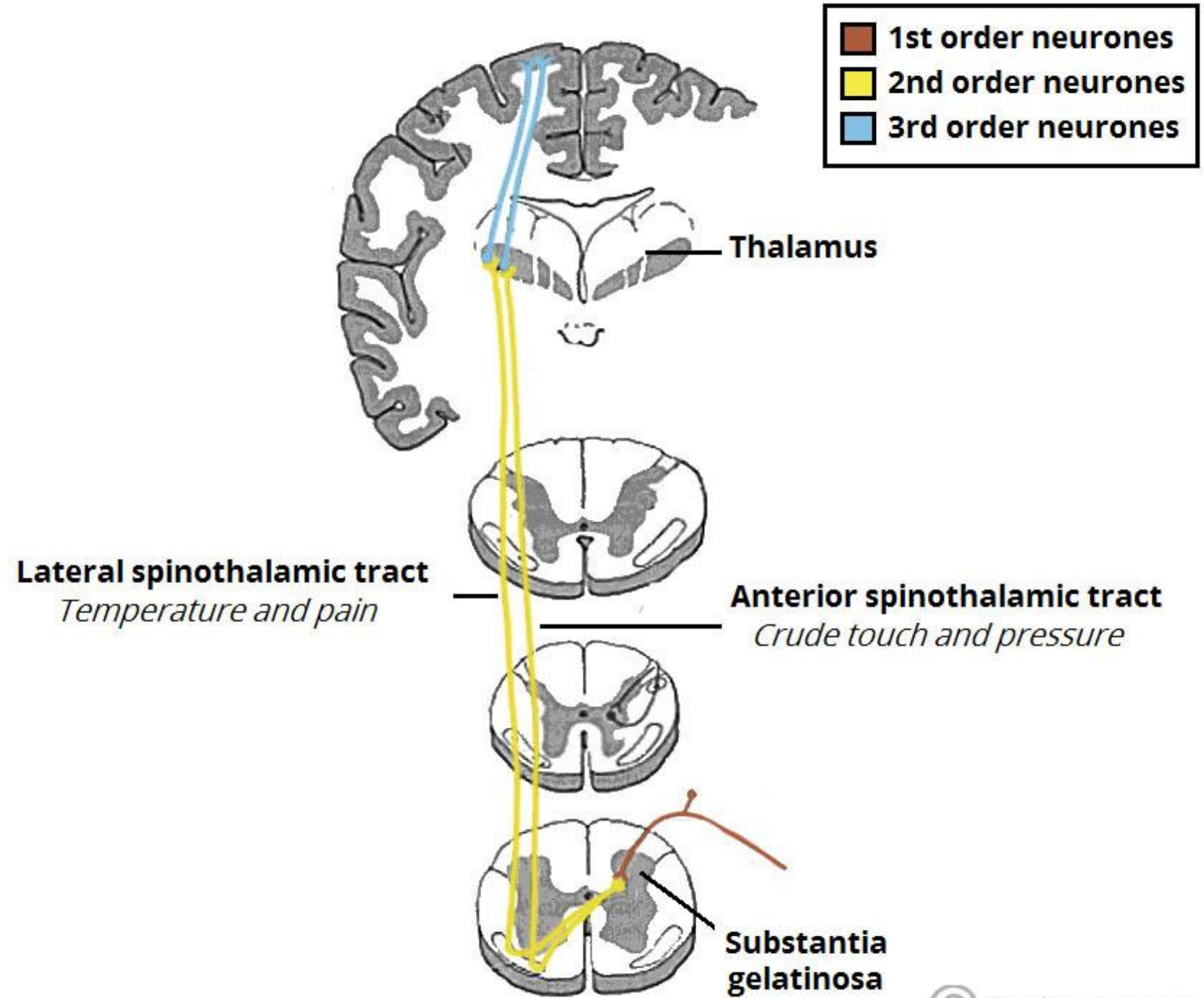


Spinal Cord Anatomy Spinothalamic Tracts

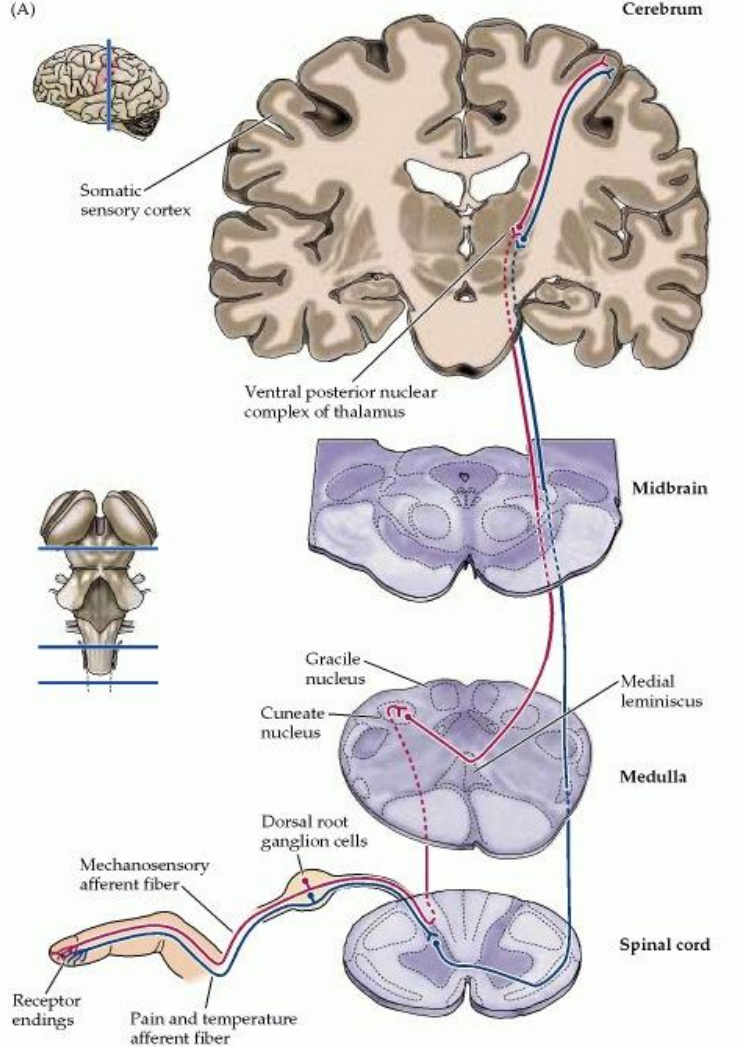


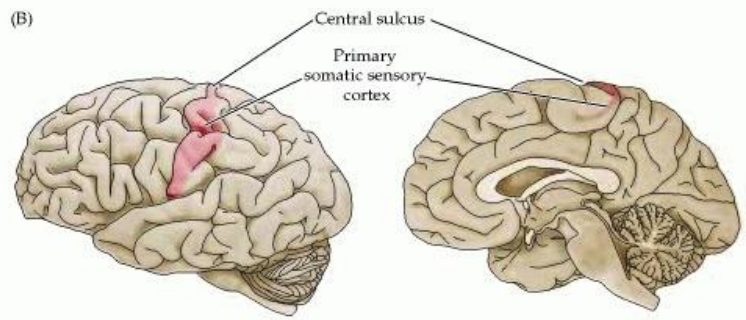


Spinothalamic tract. The central pathways for pain, temperature, touch and pressure are illustrated.

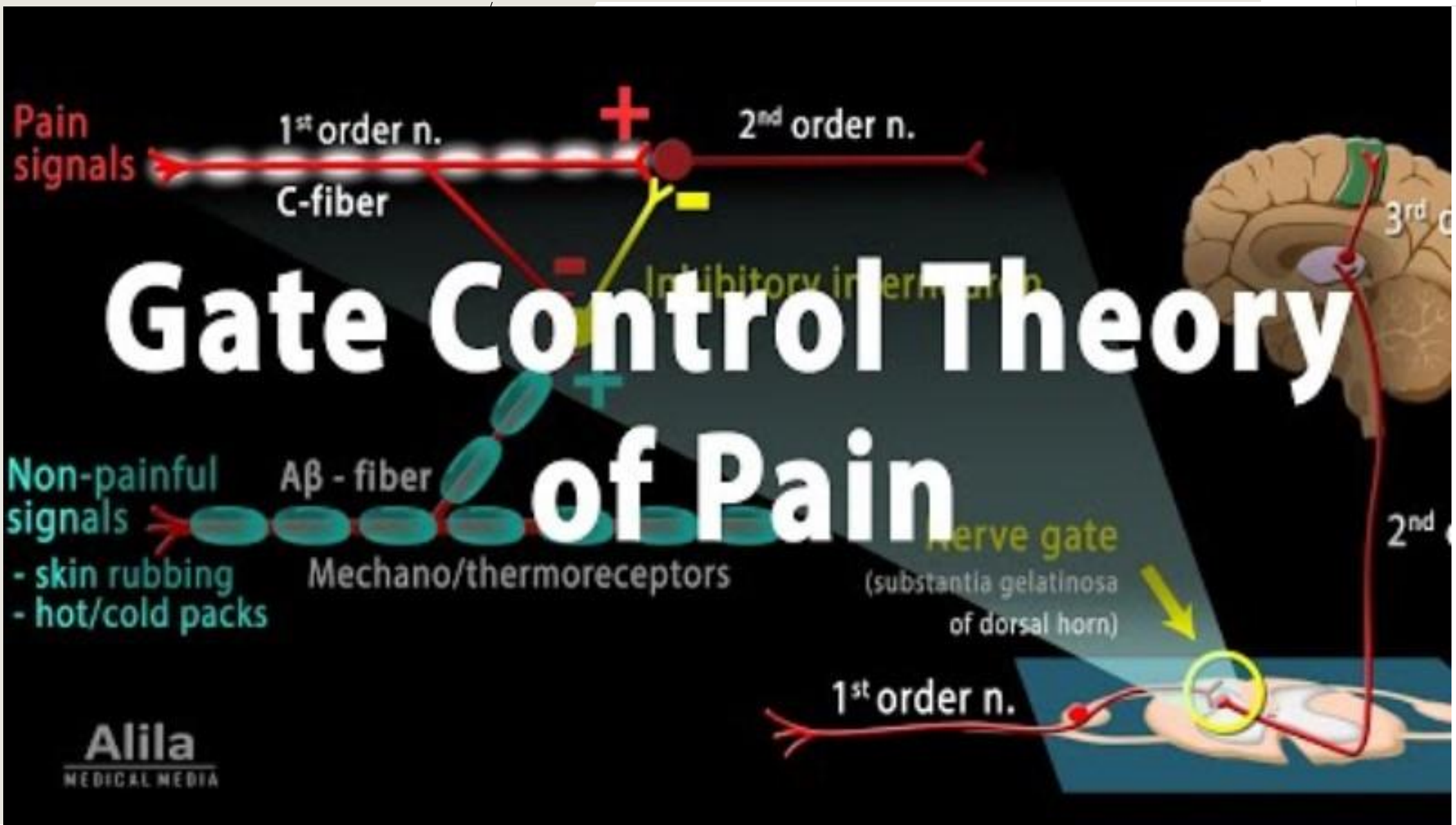


© TeachMeAnatomy


(A)
Cerebrum
Somatic sensory cortex
Ventral posterior nuclear complex of thalamus
Midbrain
Gracile nucleus
Cuneate nucleus
Medial leminiscus
Medulla
Dorsal root ganglion cells
Mechanosensory afferent fiber
Spinal cord
Receptor endings
Pain and temperature afferent fiber


(B)
Central sulcus
Primary somatic sensory cortex

Paingate Theory



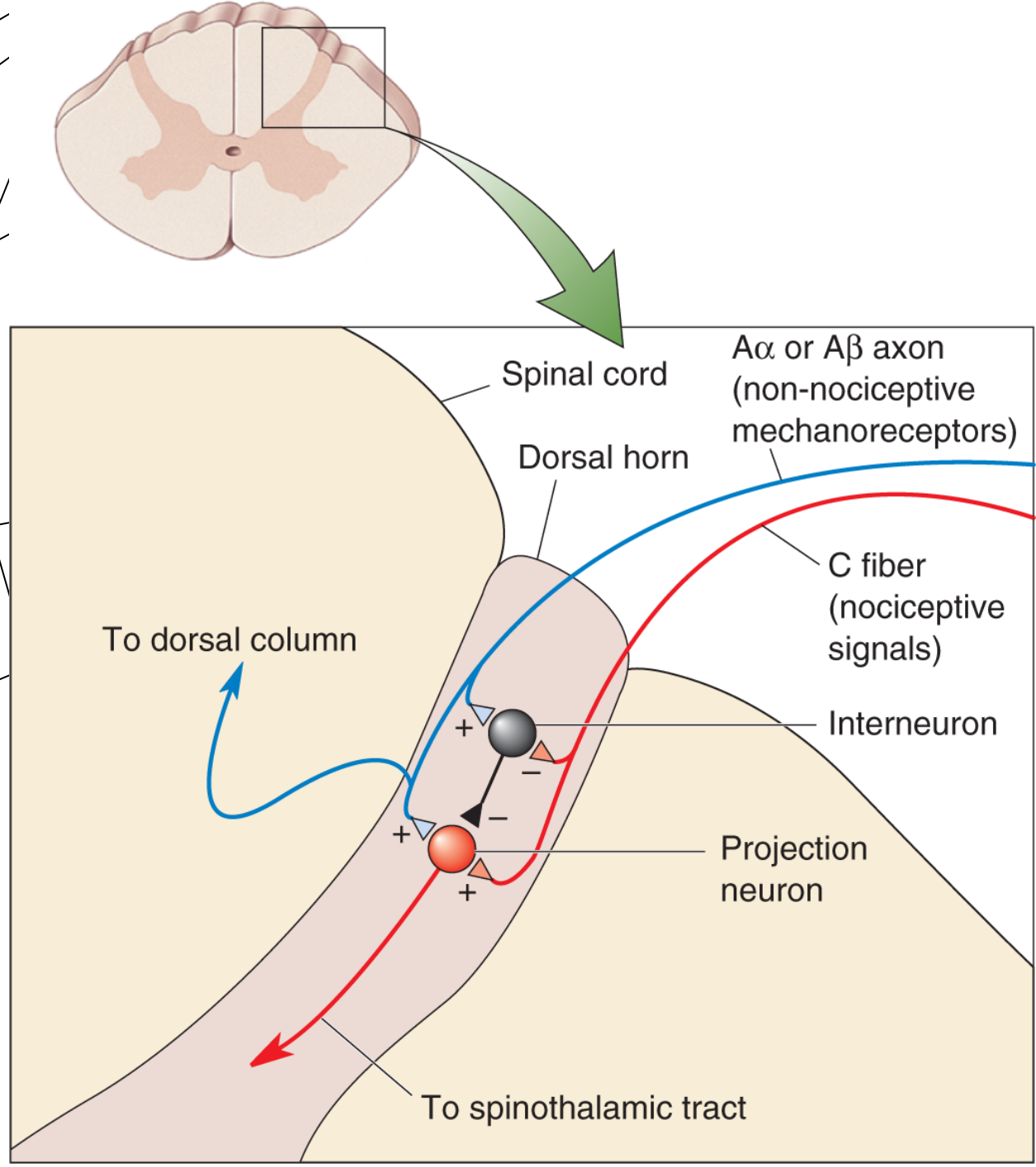


FIGURE 12.33 Melzack's and Wall's gate theory of pain. The relay of nociceptive signals by the projection neuron is gated by the activity of an inhibitory interneuron. Activity in the non-nociceptive mechanoreceptor can suppress, or close the "gate" on, nociceptive signals before they can proceed to the spinothalamic tract. The + signs indicate excitatory synapses and the – signs indicate inhibitory synapses.

Pain signals can be interrupted in the substantia gelitanosa of the spinal cord, which acts as a "gate".



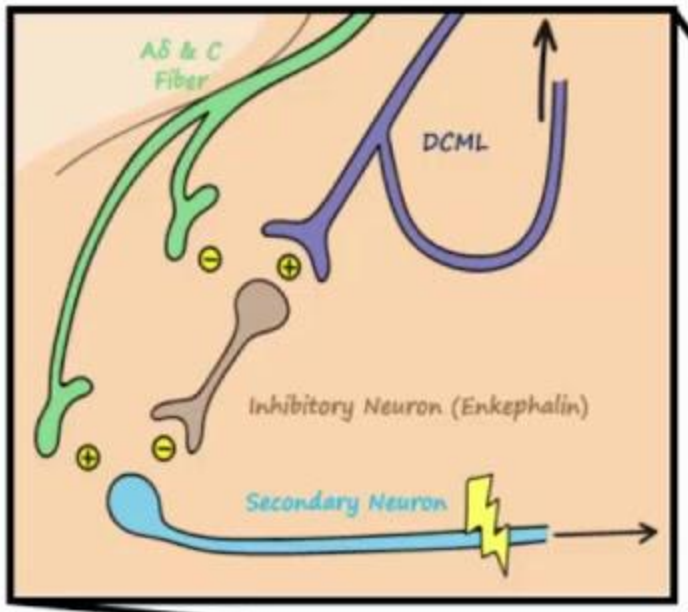
Deep touch



Painful stimulus

Pacinian corpuscle

Substantia Gelitanosa



Tertiary Neuron

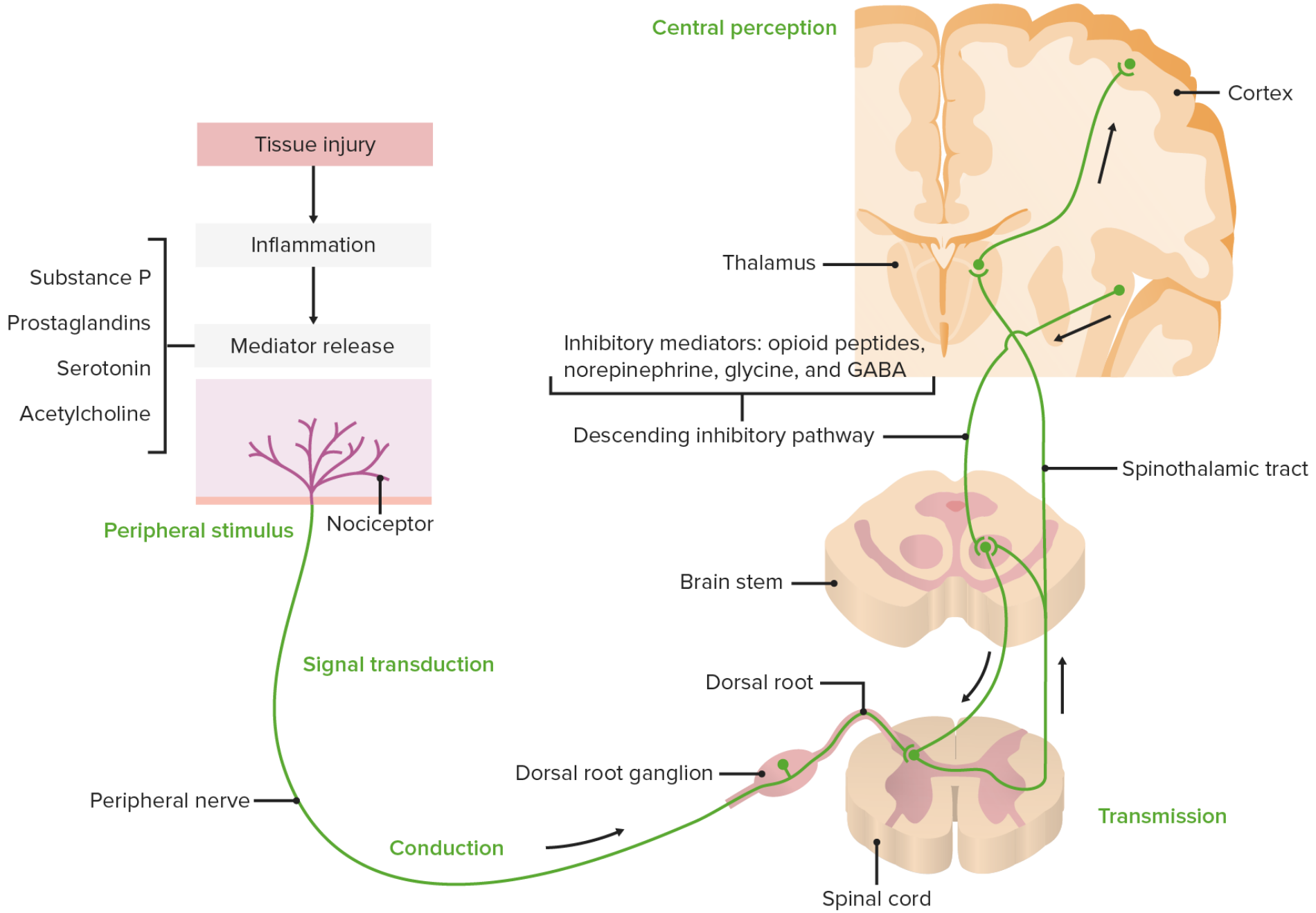
Secondary Neuron

Aδ and C Fibers

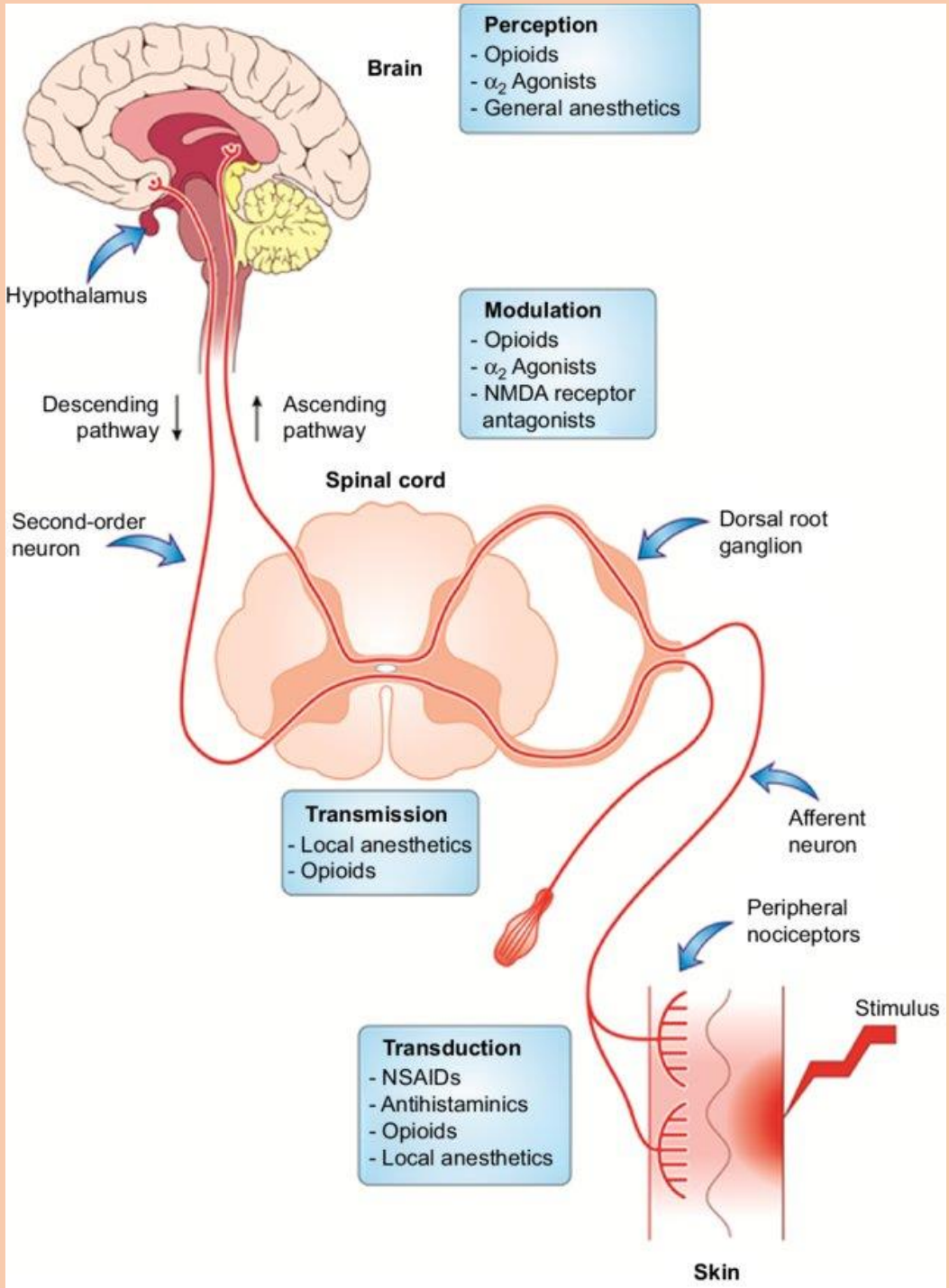
DCML

Primary Neuron

Spinal Cord


Central perception
Cortex
Tissue injury
Inflammation
Substance P
Prostaglandins
Serotonin
Acetylcholine
Mediator release
Thalamus
Inhibitory mediators: opioid peptides, norepinephrine, glycine, and GABA
Descending inhibitory pathway
Spinothalamic tract
Peripheral stimulus
Nociceptor
Brain stem
Signal transduction
Dorsal root
Dorsal root ganglion
Peripheral nerve
Transmission
Conduction
Spinal cord

Level and effect of Medications



EFFECTS OF MANUAL THERAPY

The theory is that, when we massage or articulate an area, we are activating large myelinated fibers that cause a dampening effect on the neurological C- pain fibres.

This changes the pain threshold of the nervous system and a ‘closing’ of the gating system.

Another theoretical part of the pain gate theory is the descending pathway inhibition via the periaqueductal grey (PAG). This suggests that stimulation of the periaqueductal grey matter of the midbrain causes a response in the descending pathways to the dorsal horn of the spinal cord that stimulates inhibiting interconnecting neurons situated in the substantia gelatinosa.

Bialosky, J.E., Beneciuk, J.M., Bishop, M.D., Coronado, R.A., Penza, C.W., Simon, C.B. and George, S.Z., 2018. Unraveling the mechanisms of manual therapy: modeling an approach. journal of orthopaedic & sports physical therapy, 48(1), pp.8-18.

Bialosky, J.E., Bishop, M.D., Price, D.D., Robinson, M.E. and George, S.Z., 2009. The mechanisms of manual therapy in the treatment of musculoskeletal pain: a comprehensive model. Manual therapy, 14(5), pp.531-538.

EFFECTS OF MANUAL THERAPY

Linking this to manual therapy:

Bialosky, J.E., Beneciuk, J.M., Bishop, M.D., Coronado, R.A., Penza, C.W., Simon, C.B. and George, S.Z., 2018. Unraveling the mechanisms of manual therapy: modeling an approach. journal of orthopaedic & sports physical therapy, 48(1), pp.8-18.

Bialosky, J.E., Bishop, M.D., Price, D.D., Robinson, M.E. and George, S.Z., 2009. The mechanisms of manual therapy in the treatment of musculoskeletal pain: a comprehensive model. Manual therapy, 14(5), pp.531-538.

RACING DRIVER (SAMIR) & CO-DRIVER (VIVEK)

PAIN AS A MULTISYSTEM BODY RESPONSE TO DANGER OR DAMAGE

&

BASIC RESPONSES TO THE EXPERIENCE OF PAIN AND FACTORS THAT LEAD TO THE DEVELOPMENT OF CHRONIC PAIN.

What Is Chronic Pain?

Acute Pain: short-term pain that acts as an alarm telling us something is wrong, like breaking your leg.

Chronic Pain: persistent pain that often serves no useful purpose.

Chronic pain is usually defined as: **Pain that persists beyond the normal time that tissues take to heal following an injury.**

Deemed chronic after 3 months.




www.JCTrallying.com
HiveFunnies

RACING DRIVER (SAMIR) & CO-DRIVER (VIVEK)

- Was Samir the perfect driver / Was Vivek the perfect co-pilot?
- Was the race well controlled?
- Did the car break? (no it didn’t)

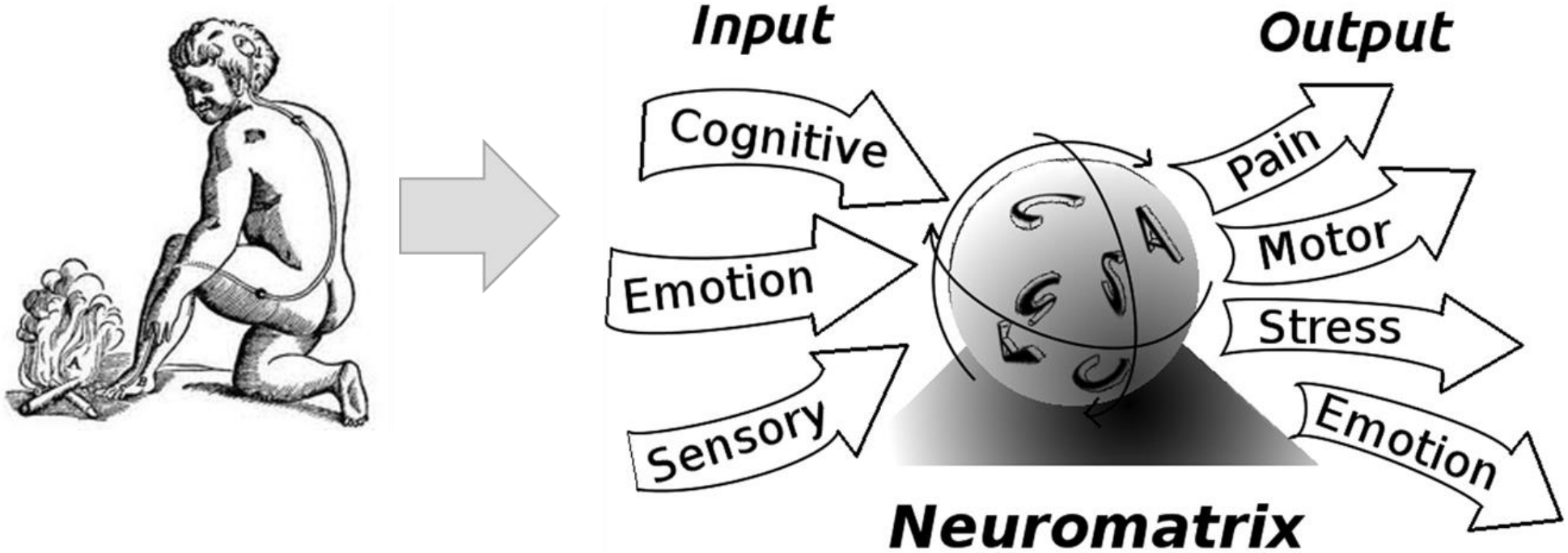
PAIN AND THE NEUROMATRIX IN THE BRAIN, MELZACK, 2001

The neuromatrix theory of pain

- **multidimensional experience with a "neurosignature" pattern**
  - **may be triggered by sensory inputs**
  - **or generated independently of them!**
- **Chronic pain syndromes, which are often characterized by severe pain associated with little or no discernible injury or pathology, remain a mystery. Pain arises from a widely distributed neural network in the brain rather than directly by sensory input evoked by injury, inflammation, or other pathology.**
- **Patterns involve activating perceptual and behavioural programs after injury, pathology, or chronic stress.**
- **Neuromatrix – genetics modified by sensory experience resulting in a neural pattern that produces pain.**

Melzack, R., 2001. Pain and the neuromatrix in the brain. Journal of dental education, 65(12), pp.1378-1382.

CULTURAL TRANSFORMATION IN THE WAY PAIN IS VIEWED, ASSESSED, AND TREATED



NAIL IN THE HEAD

NAIL IN THE HEAD

- Is it really about the nail?
- What if there was no nail, and she still has symptoms?

- What if someone has a scan showing mild osteoarthritis, but the symptoms are disproportionate to the scan?

There is a high, hard ground where practitioners can make effective use of research-based theory and technique, and there is a swampy lowland where situations are confusing "messes" incapable of technical solution.,...In the swamp are the problems of greatest human concern.

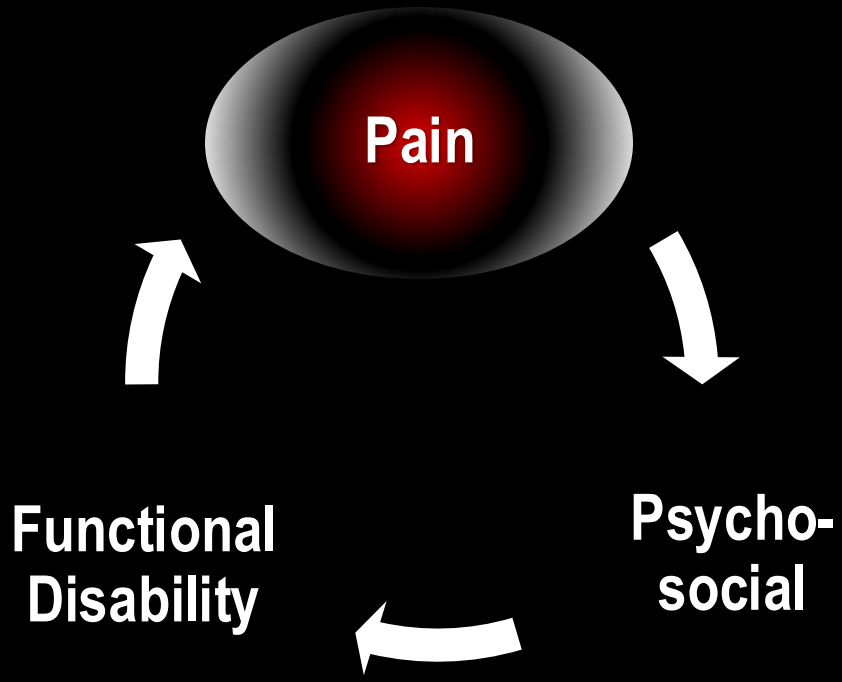
- Donald Schön -

LIBQUOTES.COM

BIOPSYCHOSOCIAL MODEL OF PAIN

PAIN

Not just a symptom
but your patient's lived experience



Pain related to damage of somatic or visceral tissue as a result of trauma or inflammation

Pain related to damage of peripheral or central nerves

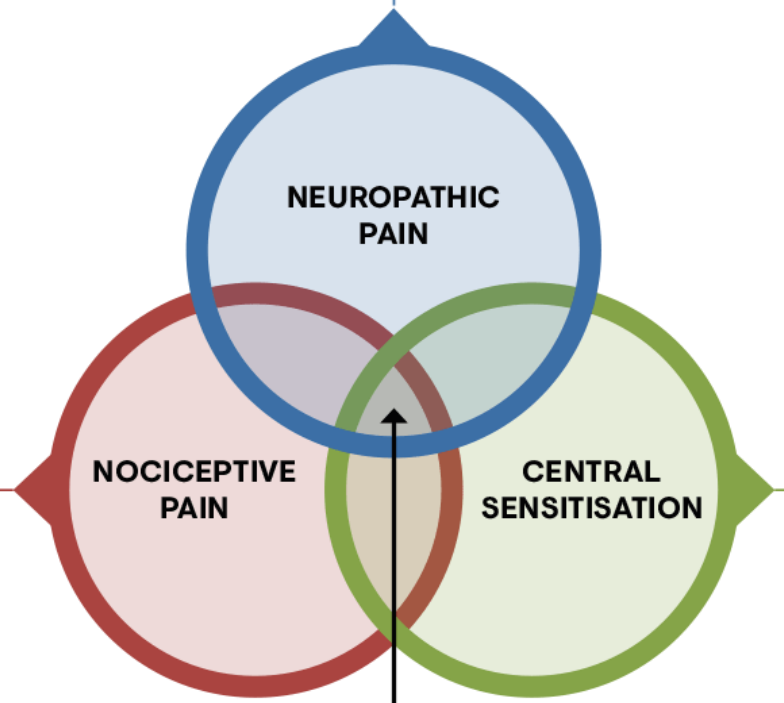
Pain without identifiable nerve or tissue damage thought to result from persistent neuronal dysregulation

NOCICEPTIVE PAIN

NEUROPATHIC PAIN

CENTRAL SENSITISATION





PREDOMINANTLY NEUROPATHIC

- PHN
- pDPN
- Lumbar or cervical radioculopathy
- Stenosis
- Tumour-related neuropathy
- Chemotherapy-induced neuropathy
- Small fibre neuropathy
- Persistent postoperative pain
- Multiple sclerosis pain
- Post-stroke pain
- Pain associated with spin cord injury

PREDOMINANTLY NOCICEPTIVE

- Osteoarthritis
- Rheumatoid arthritis
- Tendonitis, bursitis
- Ankylosing spondylitis
- Gout
- Neck and back pain with structural pathology
- Tumour-related nociceptive pain
- Sickle cell disease
- Inflammatory bowel disease

PREDOMINANTLY CENTRAL SENSITISATION

- Fibromyalgia
- Irritable bowel syndrome
- Tension-type headaches
- Interstitial cystisis/pelvic pain syndrome
- Tempo-manibular joint disorder
- Chronic fatigue syndrome
- Restless leg syndrome
- Neck and back pain without structural pathology

Mixed pain conditions with multiple pain pathophysiologies such as chronic low back pain

NOCIPLASTIC PAIN

Nociplastic pain as part of a continuum.

- Nociplastic pain as a unique category devoid of actual or threatened tissue or somatosensory damage, there is evidence for the overlap of nociceptive, neuropathic, and nociplastic pain, indicating that nociplastic pain is not a distinct entity, but part of a chronic pain continuum.

- The biopsychosocial model is particularly relevant for nociplastic pain disorders. Although often sought out, a causal factor is usually not evident. Common predisposing factors include a family history of pain, a history of pain experiences, and psychosocial factors such as psychological, emotional, sexual, or physical abuse, or a combination of these.

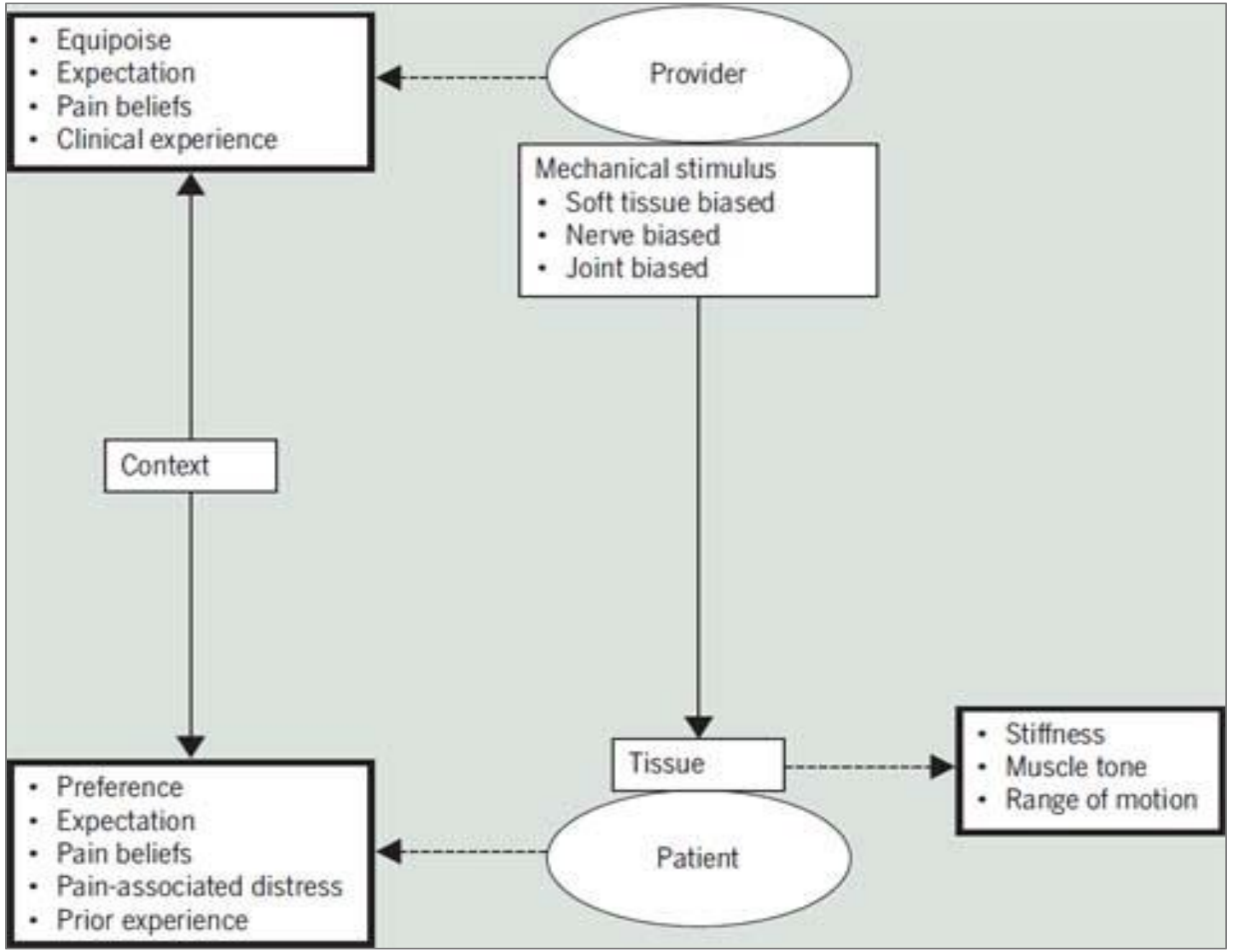


FIGURE 3. Zone 1 of the model, encompassing the interaction between the provider and patient, as well as the mechanical stimulus to the targeted tissue. Solid arrows denote a direct mediating effect. Broken arrows denote an association between a construct and its measure. Bold boxes provide examples of measurable constructs for consideration.

Journal of Orthopaedic & Sports Physical Therapy
JOSPT

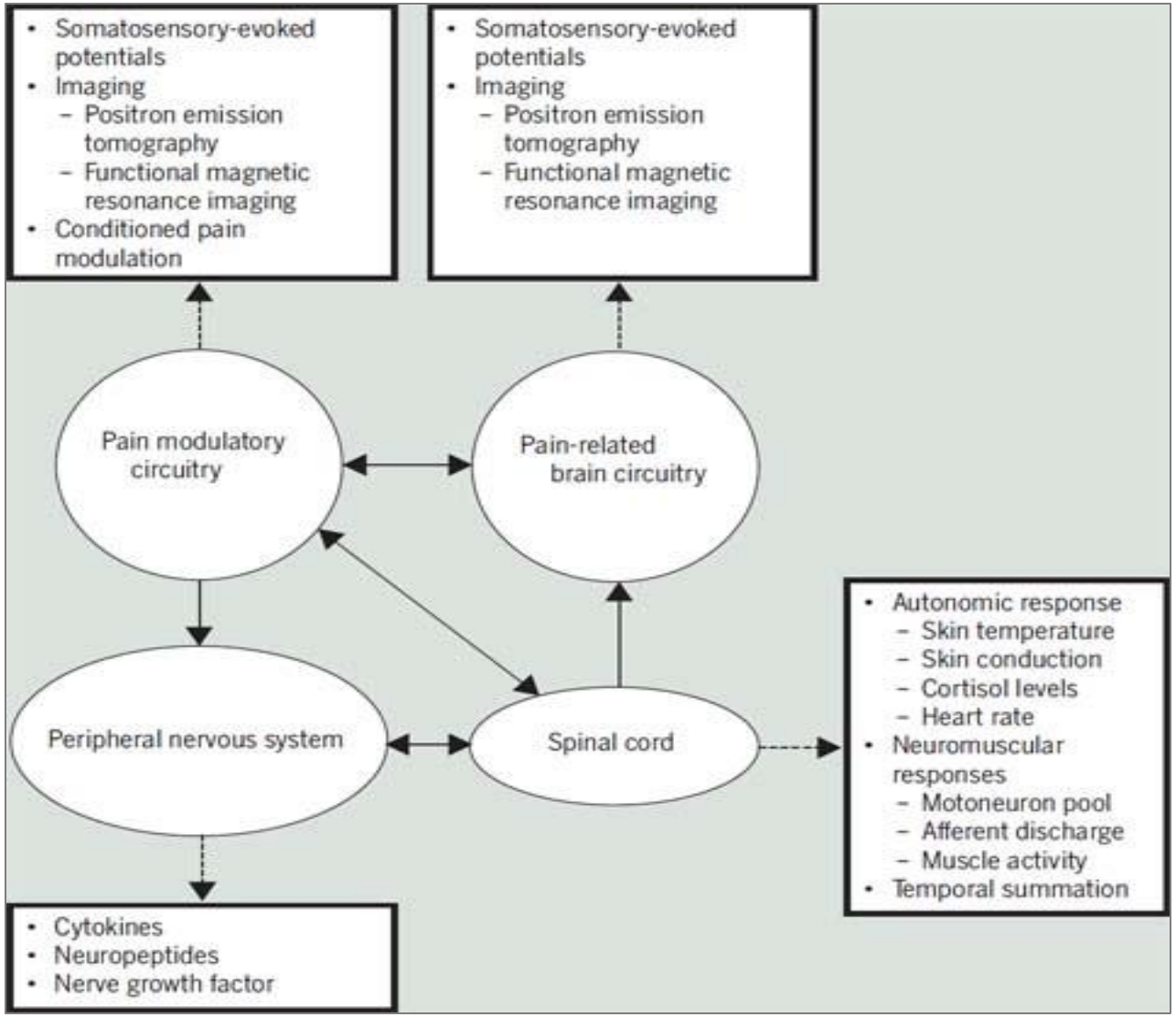


FIGURE 4. Zone 2 of the model, encompassing the nervous system response of the patient to receiving a manual therapy intervention. Solid arrows denote a direct mediating effect. Broken arrows denote an association between a construct and its measure. Bold boxes provide examples of measurable associated responses suggestive of a direct nervous system response.

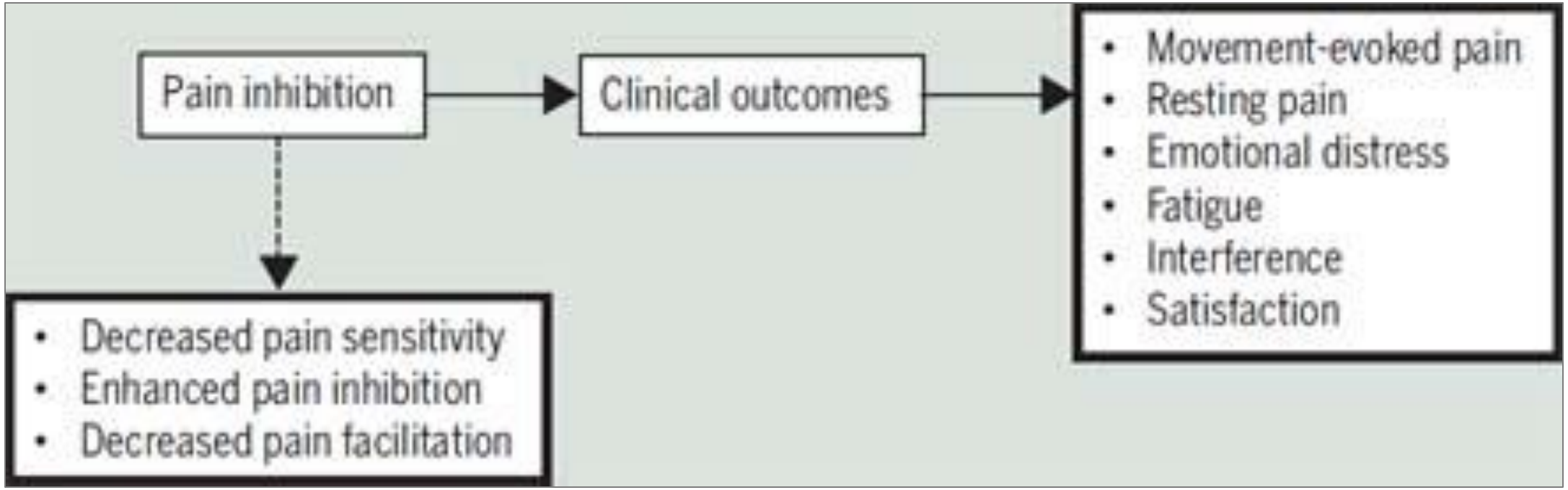
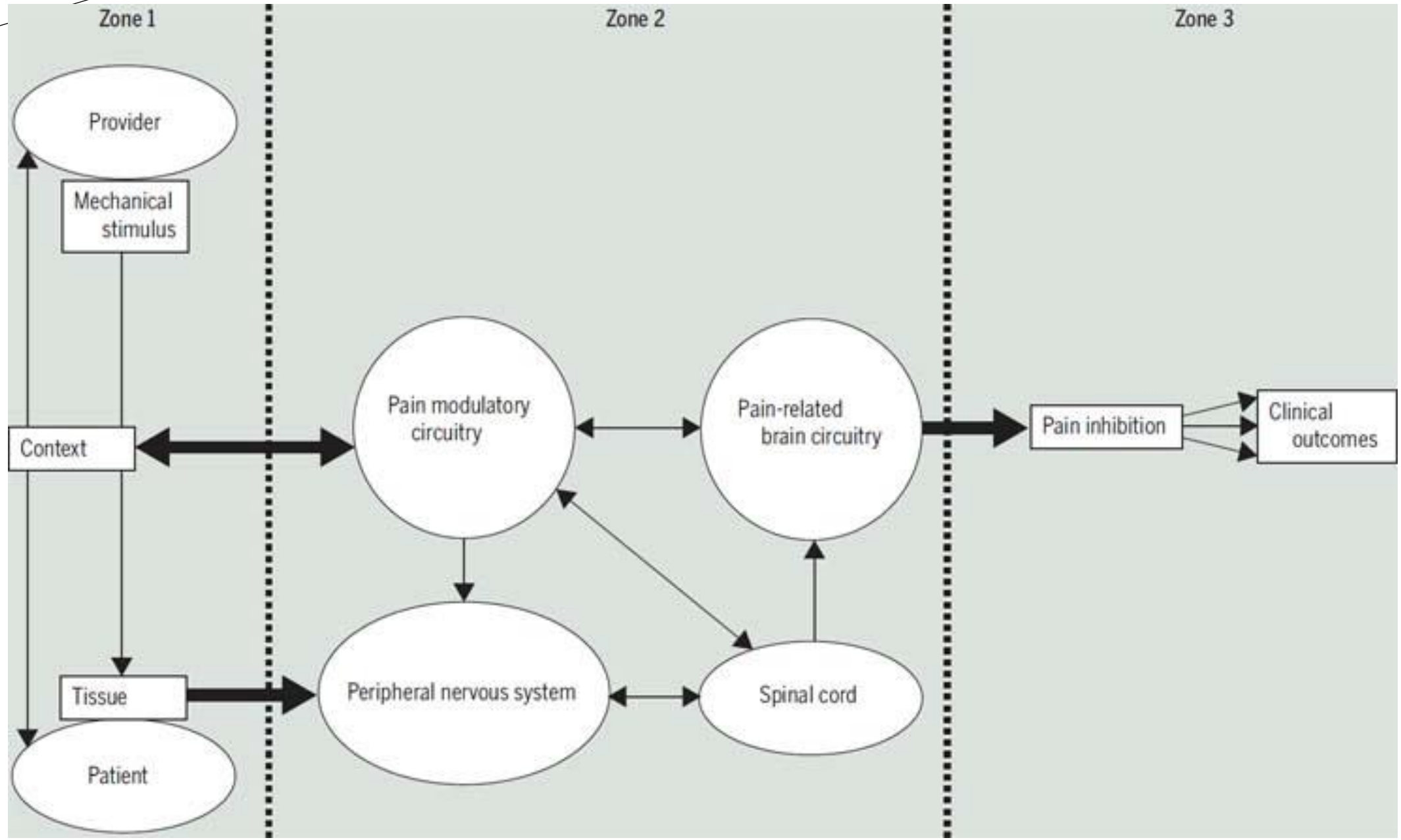
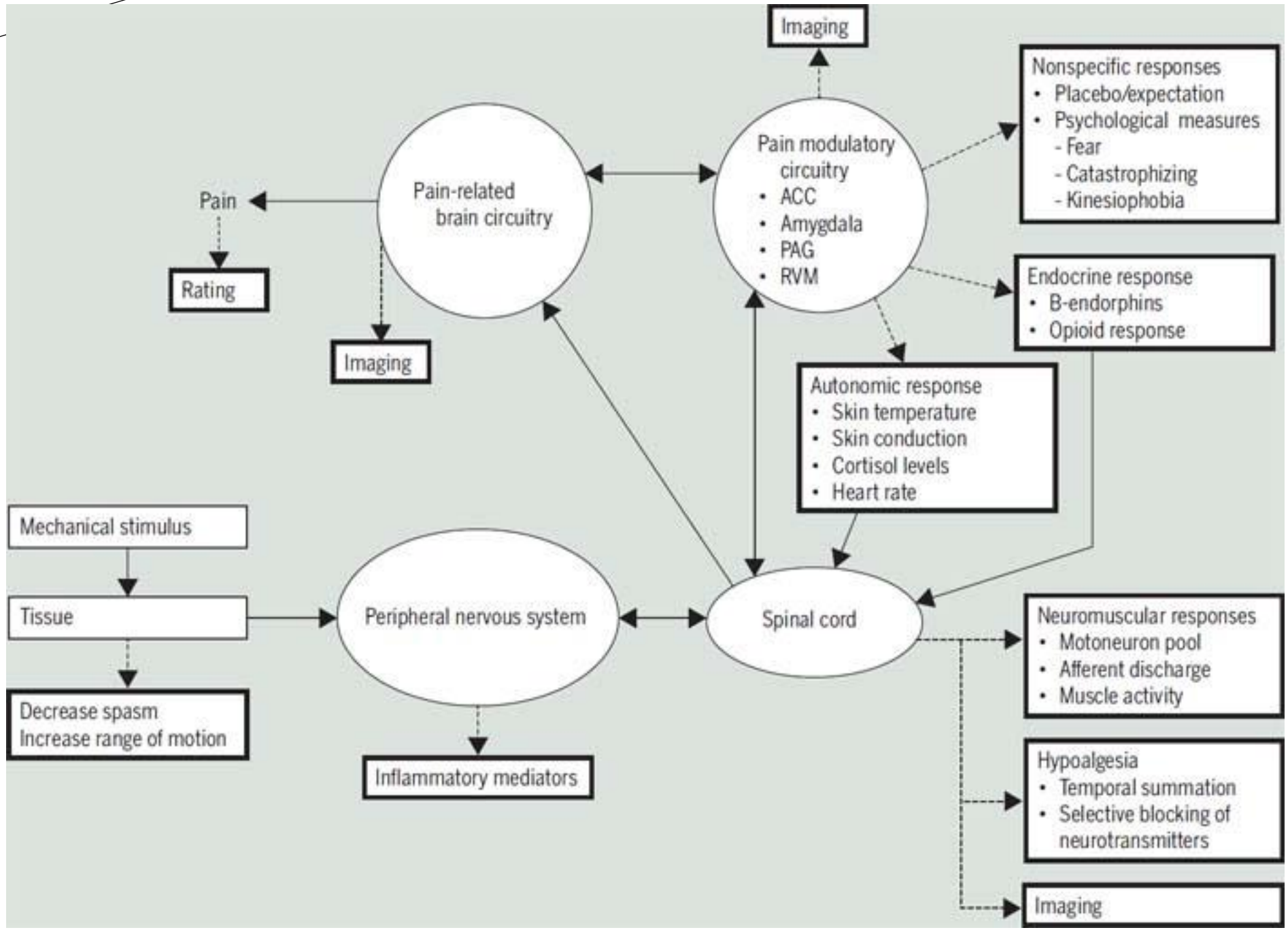


FIGURE 5. Zone 3 of the model, encompassing the outcomes of the model for which related mechanisms of manual therapy may account. Solid arrows denote a direct mediating effect. Broken arrows denote an association between a construct and its measure. Bold boxes provide examples of measurable constructs for consideration.


Zone 1
Zone 2
Zone 3
Provider
Mechanical stimulus
Context
Tissue
Patient
Pain modulatory circuitry
Pain-related brain circuitry
Peripheral nervous system
Spinal cord
Pain inhibition
Clinical outcomes


Imaging
Nonspecific responses
• Placebo/expectation
• Psychological measures
- Fear
- Catastrophizing
- Kinesiophobia
Pain
Pain-related brain circuitry
Pain modulatory circuitry
• ACC
• Amygdala
• PAG
• RVM
Rating
Imaging
Endocrine response
• B-endorphins
• Opioid response
Autonomic response
• Skin temperature
• Skin conduction
• Cortisol levels
• Heart rate
Mechanical stimulus
Tissue
Peripheral nervous system
Spinal cord
Neuromuscular responses
• Motoneuron pool
• Afferent discharge
• Muscle activity
Decrease spasm
Increase range of motion
Inflammatory mediators
Hypoalgesia
• Temporal summation
• Selective blocking of neurotransmitters
Imaging

MAO, J., 2017. CHALLENGES OF MANAGING CHRONIC PAIN. BMJ, 356.

The idea of a comprehensive pain management strategy is now more valuable than ever. Primary care clinicians, surgeons, and emergency department doctors are in the frontline of pain management, and they have a pivotal role in initiating and developing a long term strategy that includes prompt referral to pain specialists. Other key components of a comprehensive management plan include treating underlying conditions; using multiple drugs, including muscle relaxants, antidepressants, anticonvulsants, and topical agents as well as conventional analgesics, to optimise the effectiveness and minimise side effects; adding psychological interventions and physical therapy; and considering integrative approaches such as acupuncture and mind-body therapy.

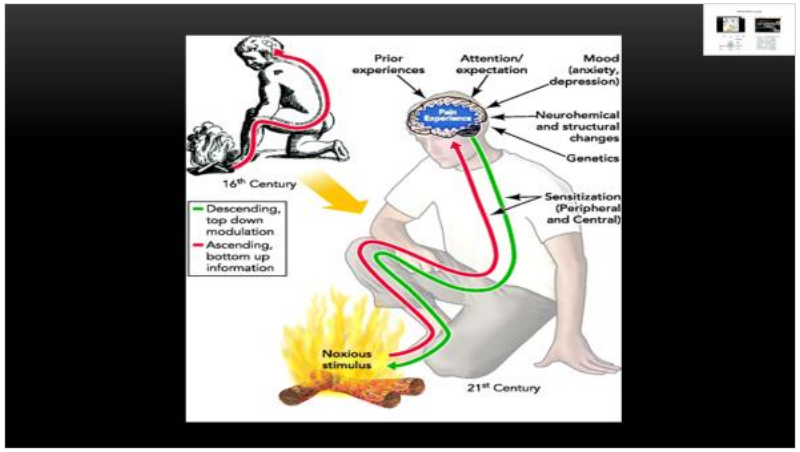
ADDITIONAL THOUGHTS

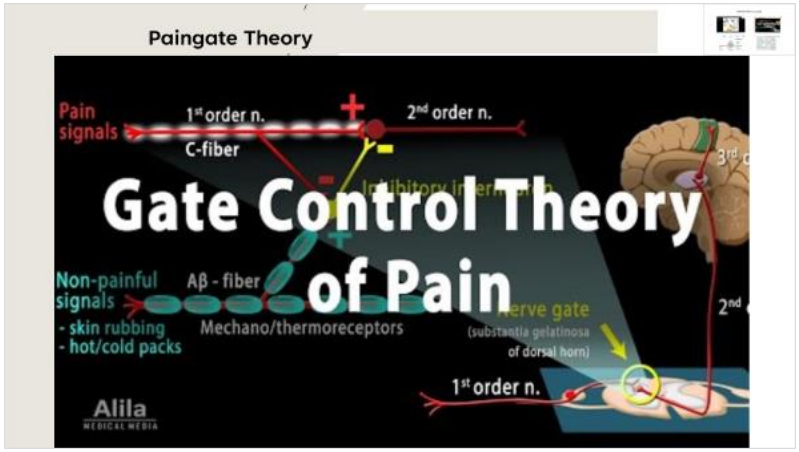
Effects of Manual Therapy?

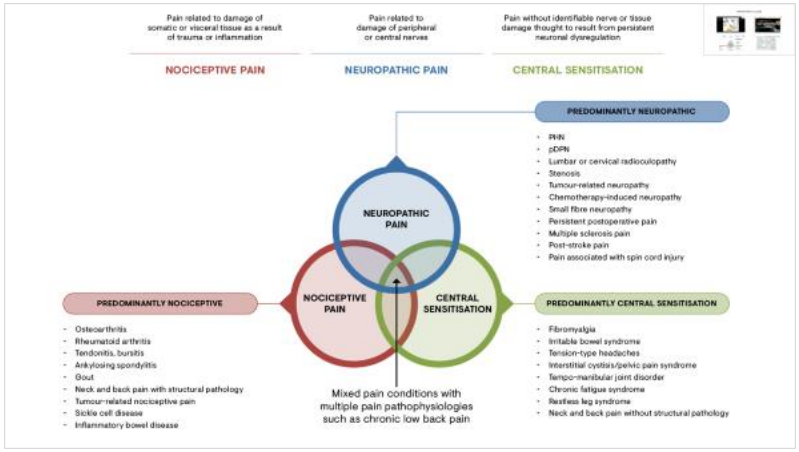
Area of growth for Physiotherapy?

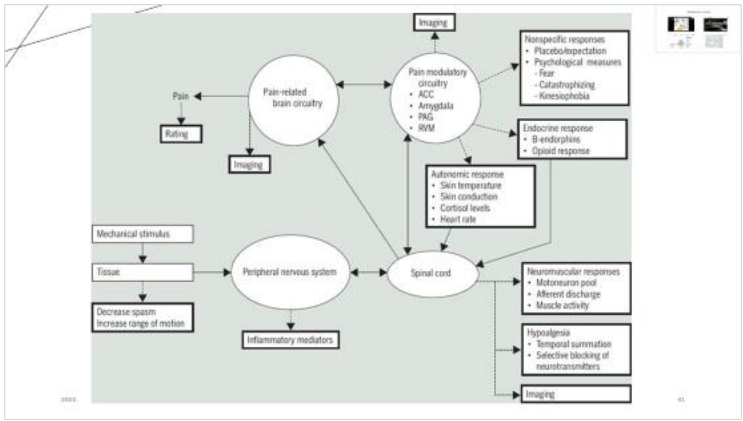
‘Re-modelling’ of brain - +ve neuroplastic changes / reversing nociplastic changes

IMPORTANT SLIDES









SUMMARY

Demonstrate an understanding of the transmission system for nociception & Describe the relationship between pain and tissue damage

- Ascending & Descending Pathways
- C-fibres, spinothalamic tract, thalamus, substantia gelatinosa, periaqueductal gray
- Neuromatrix

Multisystem body response to danger or damage & Responses to the experience of pain and factors that lead to the development of chronic pain.

- Nociceptive, neuropathic, nociplastic
- Peripheral & Central Sensitisation
- Neuromatrix patterns – remodelling.